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Evaluating Information Retrieval using Document Popularity: An implementation on MapReduce☆

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Abstract

Over the last few years, one major research direction of information retrieval includes user behaviour prediction. For that reason many models have been proposed aiming at the accurate evaluation of information retrieval systems and the best prediction of web search users' behaviour. In this paper we propose a new evaluation metric for information retrieval systems which employs two relevance factors; a relevance judgement grade and a popularity grade that represents users' vote for a document. We show that this new metric performs better than other evaluation metrics when expressing user behaviour. Moreover, in order to test the performance of our metric on different ranking algorithms, we develop a pairwise text similarity algorithm using cosine similarity which is implemented on the MapReduce model and then perform experiments on rankings generated by the algorithm.

Keywords: Information Retrieval, Evaluation, Metrics, User behaviour, MapReduce, Hadoop, Text Mining, Cosine Similarity

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☆This paper combines and extends the two papers "Reciprocal Rank using Web Page Popularity" and "CSMR: A Scalable Algorithm for Text Clustering with Cosine Similarity and MapReduce" that were presented in AIAI Workshop MHDW 2014 in Rhodes.

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1. Introduction

Information retrieval systems are increasingly evolving nowadays and as a result, evaluating them constitutes an important direction of this research area. One of the first approaches developed to evaluate a ranking system was the so called Cranfield approach [11] which is being questioned due to its low flexibility in new types of relevance data.

As a result many evaluation metrics have been developed so far aiming at the accurate prediction of user behaviour. The most conventional approach which evaluation metrics use is that of using expert judgements. Trained people - assessors are assigned to grade each document from a test collection with a five-grade scale value that represents how relevant (perfect, excellent, good, fair, bad) is a document in respect to a specific query.

Despite its wide acceptance this approach is being questioned lately mainly because it is a high-cost and infeasible process when we are dealing with huge amounts of data such as the web corpus. For that reason alternative solutions have been developed over the last years, with click-through data analysis being the most common approach [5]. Unlike expert judgements, click-through data can be collected at zero cost and in real-time [23]. What is more, click-through data reflect users' preferences than experts' opinion [15], [26].

Unfortunately click-through data analysis approaches face one major problem which lies on the proper interpretation of the retrieval quality utilization [15].

Another key issue that concerns evaluation metrics, is the user model they are based on. A user model tries to best model user interaction within the information retrieval system [10]. There are two types of user models; position models and cascade models [7]. Position models rely on the assumption that a click depends only on relevance and examination with the latter depending only on the position of the result. Cascade models on the other side, assume that users examine from top to bottom every result and after they click on a

document that they consider relevant, they terminate the search session. Some evaluation metrics relying on position models are nDCG and RBP [22], while ERR relies on the cascade model.

In this paper we introduce a novel evaluation metric which incorporates a new relevance factor called document popularity. Our evaluation metric is developed on top of the cascade model, assuming that a user clicks on a document (and stops) when he or she finds a relevant document after scanning all results from top to bottom sequentially.

Unlike previous editorial metrics we assume that the probability a user clicks on a document depends on two factors; not only on its relevance grade, but also on its popularity grade. For each document D_i, we have two values, a relevance grade as it is proposed by experts and a popularity grade as it is given by web traffic statistics. We then incorporate these two values in the click probability and evaluate how well our proposed metric captures user behaviour by comparing its performance with the performance of click-metrics. Furthermore, we examine the performance of our evaluation metric in situations where relevance judgements are not available for a proportion of the collection.

In order to evaluate the performance of our proposed metric, we perform experiments on real user search data, employing different relevance ranking algorithms. The standard relevance ranking algorithm we use is based on inference networks and is part of the Indri search engine [27]. Except from the ranking algorithm that Indri uses, we employ a second relevance ranking algorithm which is scalable for big data in order to examine how our proposed evaluation metric performs on different ranking algorithms. This algorithm uses cosine distance as a relevance similarity between documents and queries and is designed to be scalable for great number of documents. This is achieved by the use of the MapReduce component of the Hadoop Framework. This method employs the parallel and distributed algorithm implemented in Lin [18], and extends it by using the cosine distance as document similarity. Initially, the terms of each document are counted, then texts are normalized with the use of tf-idf scores and finally cosine similarity of each document pair is calculated.

The goal of this paper is to investigate the evaluation process in cases where the number of documents is extremely large. That is for example, when we have to produce a relevance ranking of 10 documents that are more similar to a specific query out of thousands of documents, and we want to examine whether the evaluation of these ranking still reflects user behavior. For that reason, we employ a scalable parallel algorithm and we did not employ any other existing relevance algorithms such as BM25.

Furthermore, in our experiments we have not employed any page importance ranking models such as PageRank [2], BrowseRank [19] or HITS [17]. These algorithms, as part of link analysis, compute the importance of a web page based on incoming or out-coming links, authority pages or other user search actions. Here in this work, we introduce the notion of page popularity which is defined in a complete different way from page importance and thus we avoid using these ranking models in order to avoid any fake bias in the evaluation process.

The rest of this paper is organized as follows. In Section 2 we refer to the recent related work in the research area of information systems evaluation and in Section 3 we present our proposed evaluation metric. In Section 4 we analyze our methodology for measuring document similarity on massive data. In Section 5 we provide evidence of the well performance of our metric and we conclude our paper in Section 6 by presenting some final remarks and future directions.

2. Related work

The evaluation of information retrieval systems has always been an important issue concerning both academic and industrial research. Consequently, many scientific works have focused on the evaluation of information retrieval systems and how to best model user behaviour, resulting in a great number of evaluation metrics such as Average Precision, Discounted Cumulative Gain (DCG) by Järvelin and Kekäläinen [16], Expected Reciprocal Rank (ERR) by Chapelle et al. [6], Expected Browsing Utility (EBU) by Yilmaz et al. [30],

e.t.c.

While the aforementioned evaluation metrics use as relevance factor expert judgements, there has been a lot of controversy surrounding the cost and feasibility of this approach [5]. As a result, several research directions have been developed including the construction of low-cost evaluation test sets, but also evaluation of test collections in cases where expert judgements are absent.

Carterette and Allan [4] and Sanderson and Joho [25] present methods on how to build test sets for evaluation at low cost. Other research works focus especially on the problem of completeness of expert judgements. Buckley and Voorhees [3] confront this limitation by introducing a new metric called "*bpref*" which measures the effectiveness of a system on the basis of judged documents only. Moreover, Sakai proposes an alternative solution which does not need a new metric [24].

In parallel to the development of evaluation metrics, the processing of massive data in the fields of information retrieval research has also attracted a lot of interest in recent years. The measurement of document similarity and as a consequence the ranking generation in response to a query constitutes a bottleneck when we have to deal with huge amounts of data such as the web corpus.

Elsayed et al. [13] and Lin [18] present a method which employs the MapReduce model in order to compute pairwise document similarity in large document collections. Bin et al. [1] propose a tf-idf algorithm which uses the MapReduce model of Hadoop and is proved to perform better when applied on massive data than the traditional method. In addition, Wan et al. [29] developed an efficient method for document clustering in large collections using MapReduce. In their work a combination of tf-idf and k-means algorithm in MapReduce is presented resulting in more efficient and scalable outcomes when dealing with massive data. Another approach proposed by Zhou et al. [31] involves the parallel implementation of k-means algorithm on MapReduce for large-scale data clustering. Finally, Mihalcea et al. [21] develop a method for measuring the semantic similarity of short texts, using corpus-based and knowledge-based measures of similarity, while they also claim that Cosine Similarity, tf-idf as well

as other methods can be used for text similarity measuring.

3. Reciprocal Rank using Document Popularity

3.1. User model

A common approach to predict user behaviour is to develop proper click user models. Click models try to best interpret and predict users' future clicks by observing past clicks. Thus, they are considered of utmost importance when developing evaluation metrics. There have been developed many user models over the last few years, such as position models, the cascade model [10], the logistic model and the dynamic Bayesian network model [7] just to name a few.

In our proposed metric we employ the cascade user model which assumes that the user performs a linear traversal through the ranking and that the rest of the results below the clicked document are not examined at all. Particularly, when the user starts a search session, he or she first examines all documents from top to bottom. As soon the user clicks on a document and is satisfied, he or she terminates the search and all results below this document are not examined at all.

Now let R_i be the probability of examination of the i-th document. It measures the probability of a click on a result which can be interpreted as the relevance of the snippet. As showed by Craswell et al. [10] R_i can be considered to represent the attractiveness of a result. We take the above assertion one step further and introduce a new factor called document popularity which contributes in the accurate measurement of examination probability. In order to develop an evaluation metric which captures user behaviour well, one should not only rely on expert judgements, but also on other factors which capture user behaviour. Popularity, as we define it from daily page views, can be considered as users' vote for a document and more importantly a value which expresses users' behaviour.

At this point we make the assumption that the probability of clicking a particular document relies not only on its relevance, but also on its popularity. This means that R_i can be considered as a function of two factors; the relevance

grade of document i as it is assigned by experts and a popularity grade of document i as it is defined by web-page statistics in the web.

3.2. Document Popularity

The main novelty of our proposed evaluation metric lies upon its relevance factor in which we incorporate the notion of document popularity. Cho et al. [8] in their scientific work define popularity $V(p, \Delta t)$ of a page p as "the number of visits or page-views the page gets within a specific time interval Δt". Here, we employ this definition but abstract it generally as document or web page popularity.

Definition 1. We define document popularity $P(p, \Delta t)$ of a document (or web page) d as the number of page views (pv) that document gets within a specific time interval Δt.

Document popularity as we define it represents users' vote for a document and can be considered as an indicator of relevance for the document from users' perspective. Thus, by incorporating this factor of relevance along with experts' judgements we could get a balanced notion of relevance for each document and not only an opinion from a trained assessor.

Hence, based on definition 1, we construct a popularity grade in accordance with the relevance grade, so that the popularity of each link can be measured and compared properly. Particularly, given a number of page views pv for a document link u, we define its popularity grade as follows:

$$p_u = \left\lfloor \frac{\ln pv_u}{5} \right\rfloor \tag{1}$$

Equation (1) was developed based on traffic statistics about every web page. More specifically, page view values pv ranged from 0 (that is no page views at all) to around 500.000.000 (Google page views) per day. In order to incorporate these large numbers in a metric we decided to get the natural logarithm from each pv value so that we don't lose the amount of information for each website.

Table 1: Popularity grade for four sample websites.

WebSite	Daily Page views	Popularity Grade
http://google.com	584.640.000	4
http://wikipedia.com	30.451.680	3
http://ceid.upatras.gr	11.228	1
http://sample-site.wordpress.com	11	0

Moreover we mapped each natural logarithm to a 5-scale ranging from 0 to 4 in order to create a grade-scale similar to that of the relevance grades.

According to equation (1) each website received a 5-scaled grade (unknown, well-known, popular, very popular and famous) according to its daily traffic statistics. Table 1 shows the popularity grade of some websites according to their daily page views.

3.3. Proposed Metric

After explaining the notion of document popularity in the previous section, we now introduce our novel evaluation metric. As stated before, this new evaluation metric employs the cascade user model which assumes that a user examines all results sequentially and stops when she finds a click on a relevant document. Our contribution in this work is the development of a new metric which incorporates a second factor of relevance called document popularity. Let g_i be the relevance grade of document i and p_i be the popularity grade of document i as defined in subsection 3.2. Then the relevance probability is a function of two factors and can be equally defined by g_i and p_i.

$$R_i = \mathcal{R}(g_i, p_i), \tag{2}$$

Here f constitutes a mapping function from relevance and popularity grades to probability if relevance. One common way to define f is as follows:

$$f(r) = \frac{2^r - 1}{2^{r_{max}}}, \tag{3}$$

where

$$r = \frac{g+p}{2}, \qquad r \in \{0, ..., r_{max}\}. \tag{4}$$

Equation (4) shows that the probability of relevance of a document can be equally defined by two factors; its relevance grade and its popularity grade. This is a well-balanced way to define relevance as there is an equal contribution of each factor. Thus, our evaluation metric does not reflect only experts' opinion, but also users' opinion for a specific document and as a result captures user behaviour better.

Lets explain our assertion by considering the following scenario. Assume a user examines all results from top to bottom after posing a query to a search engine and he or she finds two results (regardless of their position) he or she considers very relevance. If the first result's url is far more popular than the other, he or she will prefer to click the popular one. In other words, when a document is fairly relevant ($g = 1$), but very popular ($p = 4$), the probability that a user will click on it is much higher than if the document was not popular.

In order to complete the definition of our evaluation metric, we must determine a utility function. A utility function φ, takes as argument the position of each document and should satisfy that as position becomes grater, the utility function converges from 1 to 0. That is, $\varphi(1) = 1$ and $\varphi(r) \to 0$ as r goes to $+\infty$. Here we select our utility function to be $\varphi(r) = 1/r$ and thus our metric can be defined as follows:

$$RRP = \sum_{r=1}^{n} \frac{1}{r} \prod_{i=1}^{r-1} (1 - R_i) R_r. \tag{5}$$

4. A ranking algorithm implemented on MapReduce

4.1. Using cosine similarity on MapReduce

In our effort to examine our information retrieval metric's performance on different systems, we employ a text similarity algorithm based on the MapReduce model in order to produce rankings using the TREC Web Tracks dataset

of documents and queries. We then evaluate the rankings produced by a non-parallel framework and the MapReduce framework.

The text similarity algorithm (we will refer to it as CSMR – Cosine Similarity using MapReduce – for the rest of the paper) constitutes a simple approach, comprising of a preprocessing step which divides the collection and computes document vectors including tokenization, stopword removal e.t.c. Thereafter follows a second step where the tf-idf value of each term is measured in a way similar to [18] and [29]. In the last step, where a similarity score between every document vector is computed, we differentiate from [18] and use the cosine similarity measure instead.

Mahout's distance package[1], provides a number of different distance measures such as, Chebyshev, Cosine, Euclidean, Mahalanobis, Manhattan, Minkowski, Tanimoto and variations of all the above. According to Manning et al. [20] the standard way of quantifying the similarity between two documents is to compute the cosine similarity of their vector representations. Also, cosine distance is widely used for text similarity purposes and provides highly accurate results in several domains [20]. In a set of experiments we performed to investigate which distance measure to use, cosine distance metric performed better than other distance metrics in terms of precision. This is the reason we decide to employ cosine distance as similarity measure for our ranking algorithm.

The cosine similarity between two vectors A and B is computed by the inner-product of the two vectors divided by the ℓ_2-norms of the vectors:

$$cos(A, B) = \frac{AB}{\|A\|\|B\|} \tag{6}$$

When the cosine value appears to be 1, this indicates that the two documents are identical, while in cases where it is 0 (i.e., their document vectors are orthogonal to each other), there is nothing in common between them . The attribute vectors A and B are usually the term frequency vectors of the documents.

[1] org.apache.mahout.common.distance/.

Algorithm 1 presents the Mapper and the Reducer procedures as they are implemented on the MapReduce framework. In the Map phase, every potential combination of the input documents is generated and the document IDs for the key as well as the tf-idf vectors for the value is set. Within the Reduce phase, the cosine score for each document pair is calculated and the similarity matrix is also provided. Figure 1 gives a graphical representation of Algorithm 1.

Algorithm 1 Cosine Similarity

```
1:  class Mapper
2:    method Map(docs)
3:    n = docs.length
4:
5:    for i = 0 to docs.length
6:      for j = i + 1 to docs.length
7:        write((docs[i].id, docs[j].id), (docs[i].tfidf, docs[j].tfidf))
8:  class Reducer
9:    method Reduce((docId_A, docId_B), (docA.tfidf, docB.tfidf))
10:   A = docA.tfidf
11:   B = docB.tfidf
12:    cosine = sum(AB)/(sqrt(sum(A2))sqrt(sum(B2)))
13: return ((docId_A, docId_B), cosine)
```

4.2. Implementation details on MapReduce and Hadoop

Hadoop software library is a framework developed by Apache, suitable for scalable, distributed computing. It allows storage and large-scale data processing across clusters of commodity servers. The innovative aspect of Hadoop is that there is no absolute necessity of expensive, high-end hardware. Instead, it enables distributed parallel processing of massive amounts of data on industry-standard servers with high scalability for both data storing and processing. Therefore it is considered to be one of the most popular frameworks for big data analytics and is applied in a wide range of engineering [28] and infor-

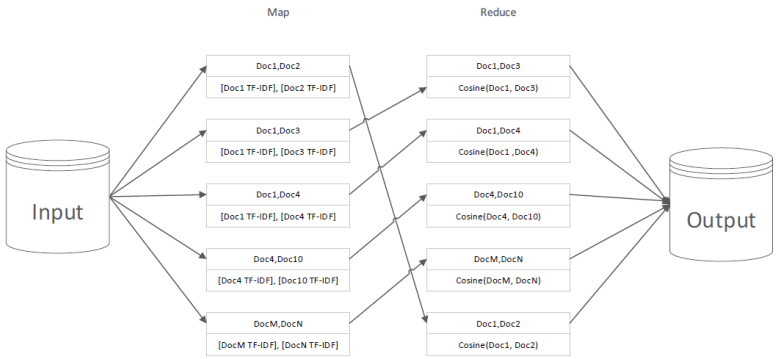

Figure 1: Diagram for algorithm 1.

mation sciences, including information retrieval. Particularly, Hadoop has two main subprojects: HDFS (Hadoop Distributed File System) and MapReduce.

MapReduce is the main component of Hadoop. It is a programming model that allows massive data processing across thousands of servers in a Hadoop cluster. The MapReduce paradigm is derived from the Map and Reduce functions of the Functional Programming model. A MapReduce program consists of the Mappers and the Reducers. In the Map phase the master node divides the input into smaller partitions and distributes them to the worker nodes. Then a worker node may repeat the same step recursively. As soon as this procedure is completed, the master node collects the key-value pairs resulted from the Mappers and distributes them to the Combiners to combine the pairs with the same key. This phase is known as the Shuffle and Sort phase. Finally, the keyvalue pairs are distributed to the Reducers that produce the final output. This step is called the Reduce phase. An overview of the MapReduce process is presented in Figure 2.

Regarding the implementation needs, two major tools - frameworks are used: Apache Hadoop (including HDFS and Yarn), and Apache Mahout. The input data is distributed on the cluster by HDFS (Hadoop Distributed File System), that is, each document of the corpus (training set) is partitioned into chunks and each of them is saved in some nodes of the cluster. In addition, Yarn

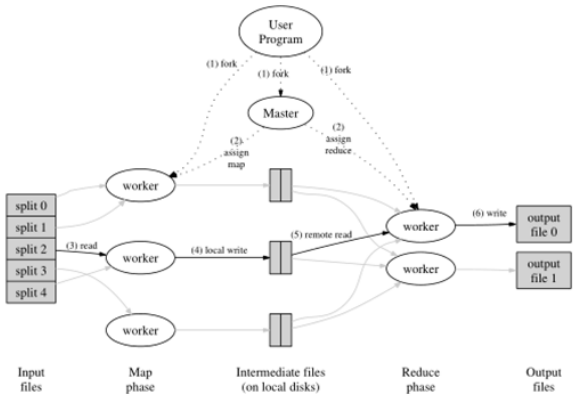

Figure 2: Overview of the MapReduce process [12].

(MapReduce 2.0) is used for MapReduce jobs execution on the cluster. Finally, we take advantage of some classes of Mahout to normalize the input documents with tf-idf scores and also, to calculate the cosine imilarity of the candidate pairs.

The proposed methodology consists of three stages: Data Preprocessing, Query Processing and Similarity Measurement. At the first stage, the whole input corpus is converted into tf-idf vectors and saved in Hadoop Sequence File Format, which is the standard Hadoop serialization file format. This is achieved simply, by using Mahout *seqdirectory* (used to serialize the documents) and *seq2sparse* (used to convert the input vectors into tf-idf vectors) bash commands.

As the input corpus is successfully preprocessed in the desirable format, each query is converted into a tf-idf vector based on the dictionary created by the corpus preprocessing. Algorithm 2 presents in Mahout pseudocode the query processing stage. Finally, in the last stage, the results are calculated in a parallel manner by using one reducer for each query, that is, the mapper takes the query id as the key and the corresponding vector as the value. Next, the reducer measures the similarity between the query and each document of the corpus.

The reducer procedure for the cosine similarity measurement for each query

Algorithm 2 Query Processing

```
class QueryTfidfReducer
  method reduce(key, value, context)
    dict = load_dict(dictionary.file − 0)
    df = load_df(part − r − 00000)
    N = number_of_documents(input)
    vectorized_query = tfidf_vectorize(query, dict, df, N)
    context.write(query_key, vectorized_query)
```

is presented in Algorithm 3 in Mahout pseudocode. The final output is a ranking of depth 5 to 20 (we choose it according to our experimental needs) documents for each query in the collection.

Algorithm 3 Query Cosine Similarity Reducer

```
class QueryCSReducer
  method reduce(key, value, context)
    corpus = load_corpus()
    similarities = hashMap(empty)
    queryId = key
    queryValue = value
    foreach doc in corpus
      sim = cosine_sim(doc, value)
      similarities.put(queryId + doc.key, sim)
return returnsimilarities.top(20)
```

5. Experimental setup and evaluation

In this section we provide evidence about the well performance of our novel metric. Evaluating information retrieval metrics is considered a difficult task mainly because "there is no ground truth to compare with". Nevertheless, a common approach involves comparison with click metrics [6], [9]. Generally, click-through data are considered as a good indicator of user's search behaviour

and are frequently used in evaluation of information retrieval systems [14]. Thus, by comparing a novel evaluation metric with metrics derived by click-through data one can see how it captures user behaviour and satisfaction.

In addition, after evaluating our new information retrieval metric we include it in a pool of metrics which will be used in order to evaluate rankings generated from our developed text similarity algorithm.

5.1. Proposed metric evaluation

In order to evaluate our novel information retrieval metric we needed click-through data for the construction of the click metrics. Due to the fact that click-logs from search engines with relevance judgements where not available to us, we used crowd sourcing to collect click-through data. Particularly, we developed a system based on Indri[2] search engine, using judged documents and queries from the TREC Web Tracks 2008-2012 collection. More specifically, the dataset we used comprised of more than one billion web pages including relevance judgments for each web page. We then used Indri to obtain rankings of depth 20, according to the relevance of each document with a specific query. The collection provided originally 200 queries related to 200 different topics. Search users from the Information Retrieval class of the Computer Engineering and Informatics Department of Patras were asked to perform an informational search on a number of predefined queries according to a special informational need.

Click-through data from each interaction with the system were collected in log files resulting in about 200k search sessions. We intersected these click through data with relevance judgements as they were defined by TREC Web Tracks on a five-grade scale ($0 \rightarrow 4$). We also intersected click through data with popularity grades which were computed from daily page views as shown in Section 3.2. Finally, each document was graded according to a five-grade

[2]http://www.lemurproject.org/. The lemur project. University of Massachusetts and Carnegie Mellon University.

scale from 0 to 4. The final dataset consisted of relevance and popularity grades for each result (document) of each query. For some queries of TREC Web Tracks though, all of their results (documents) were graded with zero relevance judgements or with zero popularity grades. We removed these queries in order to simulate a search process which reflects reality, ending with click-through data for 167 different queries.

5.1.1. Evaluation using click metrics

In the previous paragraph we mentioned that by comparing editorial metrics with click metrics we can measure how well a metric captures user behaviour. With the latter being our main goal we computed correlation between editorial metrics (including our proposed metric) and click metrics. More specifically the editorial metrics uses were:

- **Normalized Discounted Cumulative Gain**
- **Average Precision**
- **Expected Reciprocal Rank**
- **Reciprocal Rank using Document Popularity**

While the click metrics used for comparison were:

- **Precision at Lowest Rank**
- **Max, Min and Mean Reciprocal Ranks of the clicks**
- **UCTR**

Table 2 depicts the results derived from the comparison between famous editorial metrics including rep and click metrics. The values in the table are Pearson correlation scores, where higher score represents higher correlation.

As we can see in Table 2, our proposed metric shows higher correlation with click metrics than other editorial metrics. To be more specific, RRP outperforms position-based metrics such as nDCG and Average Precision and seems

Table 2: Pearson Correlation between editorial and click metrics.

	PLC	MeanRR	MinRR	MaxRR	UCTR
nDCG	0.498	0.497	0.503	0.445	-0.024
AP	0.402	0.417	0.395	0.396	-0.004
ERR	0.528	0.512	0.517	0.459	0.064
RRP	0.559	0.554	0.588	0.472	0.041

to correlate better with click metrics than cascade-based metrics such ERR. Moreover, a high correlation between ERR, RRP from editorial metrics and Mean, Max, Min RR, PLC from click metrics is showed in Table 2. This can easily explained as both ERR and RRP use as utility function the reciprocal rank $1/r$.

On the other side, an almost zero correlation between all metrics and UCTR is observed. This is mainly because UCTR does not account for clicks (rather for their absence) and therefore we can ignore its correlation scores.

Moreover, in our effort to improve the performance of our proposed metric, we altered the weights of our relevance factor in the gain function. Initially, the relevance factor r was a linear combination of gi and pi where each value contributed equally (equation (4)). A balanced contribution of expert's and user's vote does not always result in the best performance. After a number of experiments, we concluded that the ultimate performance of our metric is achieved when relevance judgements are weighted higher than popularity grades. Particularly when we have:

$$r = 0.7 \times g_i + 0.3 \times p_i, \tag{7}$$

our metric shows the highest scores. This can be explained as users are seriously affected by the popularity of a document, but the actual relevance of the document is perceived by its content. Table 3 shows the results we obtained when we apply equation (7) in the gain function of our metric. As we can observe, RRP scores even higher than in our initial approach depicted in Table

Table 3: Pearson Correlation between editorial and click metrics with optimized gain function.

	PLC	MeanRR	MinRR	MaxRR	UCTR
nDCG	0.498	0.497	0.503	0.445	-0.024
AP	0.402	0.417	0.395	0.396	-0.004
ERR	0.528	0.512	0.517	0.459	0.064
RRP	0.578	0.562	0.588	0.490	0.083

2. This means that when expert's assertion is weighted higher than documents popularity, we can predict user's behaviour more accurately.

5.1.2. Performance in incomplete collection

The continuous increasing amount of data in the web makes it more and more difficult to maintain relevance judgements for every document and as result evaluation becomes infeasible. Thus it is common place to have documents in huge collection which lack relevance judgements. For this reason various approaches have been proposed over the years on how to battle the missing judgements problem. Buckley and Voorhees [3] first proposed a preference based evaluation metric which measures the effectiveness of an information retrieval system using judged documents only, while Sakai proposes an alternative solution which does not need a new metric [24].

Our proposed metric which accommodates two factors of relevance can battle the missing relevance judgement issue in a complete different way. In situations where there exists a document with missing relevance judgement, we employ the second factor, the popularity grade which accounts for user preference. In order to enhance our assertion we conducted experiments on how correlation between click metrics and our proposed metric was affected when some queries of our dataset lacked relevance grades.

Particularly, from the initial set of 167 queries, we created 2 sets where in the first set 41 from 167 queries lacked relevance grades and in the second set 83 from 167 queries lacked relevance grades. We then computed the correlation

between click and editorial metrics of each partly "unjudged" dataset. Figures 3, 4 and 5 show the correlation diagram for the initial dataset and the two partly "unjudged" datasets. In Figure 4 where a small set of the dataset is "unjudged", our metric retains the highest scores in correlation with click metrics. As we increase the amount of "unjudged" queries in Figure 5 (half of the dataset contains no relevance grades), we observe deterioration of our metric's performance. We can thus easily conclude that in cases where the amount of relevance judgements absence is not extensive, our proposed metric can still express user behaviour better than the other editorial metrics.

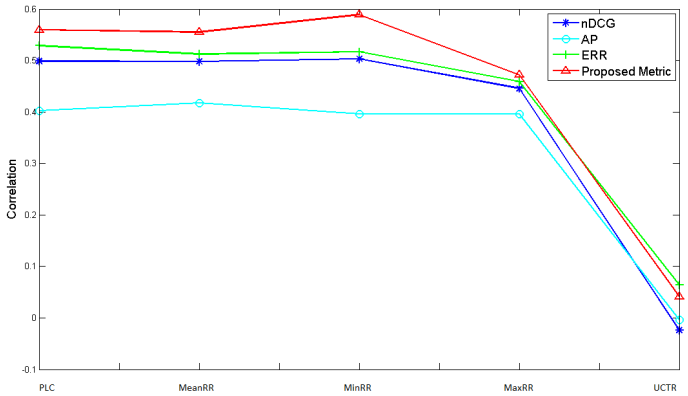

Figure 3: Correlation between Editorial and Click metrics when no queries are "unjudged".

5.2. Evaluation on MapReduce framework

In this section we perform experiments in order to examine our metric's performance on different ranking algorithms. In this set of experiments we used again the TREC Web Tracks 2008-2012 collection and we used our parallel text similarity algorithm in order to generate rankings of depth 5, 10, 15 and 20 documents. We then evaluated our proposed algorithm, with a number of evaluation metrics such as nDCG, Average Precision, ERR, including our proposed metric RRP. The results were compared with the initial rankings we obtained from a non-parallel algorithm for text similarity (Indri).

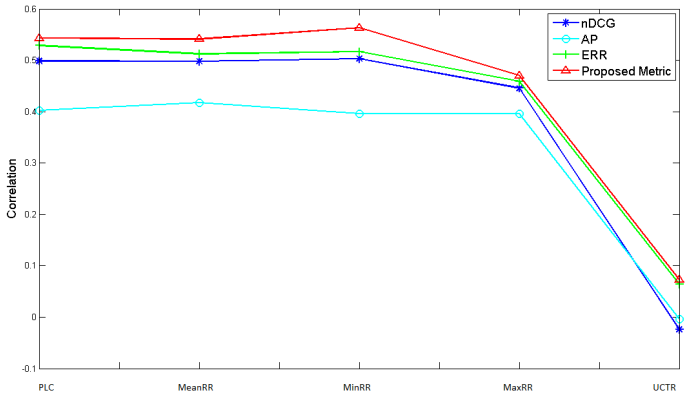

Figure 4: Correlation between Editorial and Click metrics when 41 queries are "unjudged".

Initially, our initial goal was to measure the effectiveness of the two algorithms by treating them as black boxes. That is for each query each algorithm produces an ordered set of results without taking into account any user behaviour. For that reason we used nDCG and Average Precision in a first level of evaluation, as they are used for the evaluation of information retrieval systems using test collections, query topics and relevance judgements, without taking into account any user preference.

Figures 6 and 7 present the scores of nDCG and Average Precision respectively at depths 5, 10, 15 and 20, for the two ranking algorithms. We can observe that the two algorithms show nearly identically scores in both metrics, a fact that verifies the well performance of CSMR ranking algorithm.

Now after providing evidence about the solid ranking results CSMR produces, we aim at the application of our proposed metric on rankings of CSMR and verify that RRP expresses user behaviour better than other evaluation metrics. For that reason we collect click-through data in a same way as described in Section 5.1 with the only difference that the ranking algorithm in the system now is CSMR. Then we compute the correlation scores between evaluation metrics, as they were obtained after the evaluation of CSMR rankings of depth 20, and click metrics as they were obtained from click-logs. Table 4 shows the

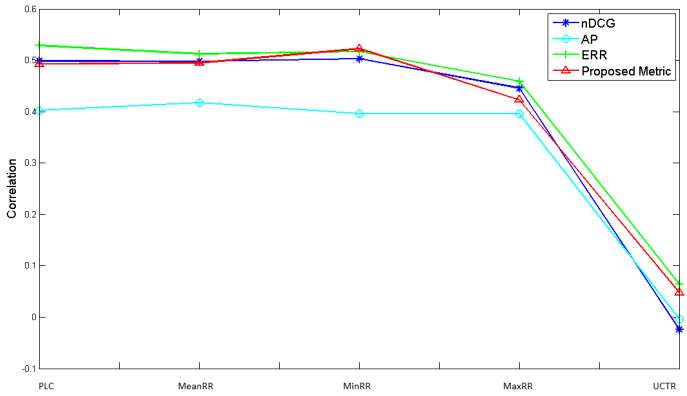

Figure 5: Correlation between Editorial and Click metrics when 83 queries are "unjudged".

Table 4: Pearson Correlation between editorial metrics applied on CSMR rankings and click metrics.

	PLC	MeanRR	MinRR	MaxRR	UCTR
nDCG	0.094	0.252	0.278	0.118	-0.094
AP	0.015	0.172	0.219	0.001	-0.052
ERR	0.081	0.242	0.214	0.124	0.053
RRP	0.171	0.370	0.312	0.308	-0.126

Pearson correlation scores of click and evaluation metrics. As we can see RRP appears to achieve better scorings, and therefore we can say that it can express user behaviour better even when it is applied to different ranking algorithms. We note here that the differences in the scores from Tables 2 and 4 hold due to the fact that we are using different ranking algorithms.

6. Discussion

In this paper we proposed a novel evaluation metric for information retrieval systems. This novel evaluation metric, predicts user behaviour more accurately by the use of a novel relevance factor called popularity grade, which is derived from the amount of page-views a document gets within a time interval. The

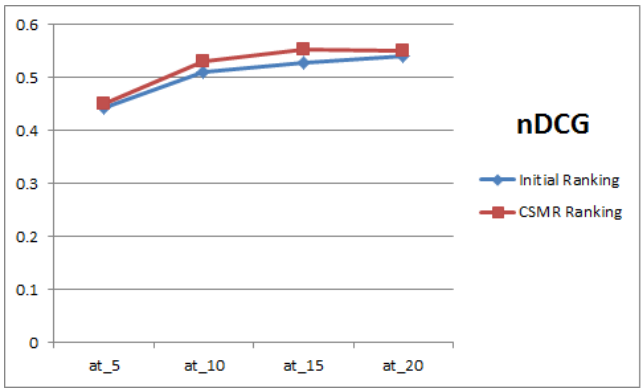

Figure 6: nDCG scores for rankings at depth 5, 10, 15 and 20 for CSMR and our initial ranking algorithm (Indri).

number of page-views of a document can be seen as users' vote for this document and hence by the combination with experts' opinion we obtain a well-balancing which results in an evaluation metric that captures user behaviour more accurately. We conducted experiments on a dataset with judged documents from TREC Web Tracks 2008-2012 and click-through data from students' search sessions. Results showed that our metric performs better than other evaluation metrics. Moreover we investigated different linear combinations on the gain function of our metric and improved it by adding more weight on the relevance grade of a document, rather on its popularity grade. That led to an even better performance of our metric. As an extension we investigated how our evaluation metric performs in situations where relevance judgements for some documents are missing. Again our metric showed better performance compared to other metrics when less than the half of the total number of expert judgements are missing. After providing evidence about our metric's well performance, we apply our metric along with other well-known evaluation metrics on rankings generated by a parallel text similarity algorithm implemented on MapReduce. We implement CSMR using the Hadoop project and verify its well performance by evaluating it with a number of information retrieval metrics including our proposed metric. Again RRP appears to express user behaviour better than other

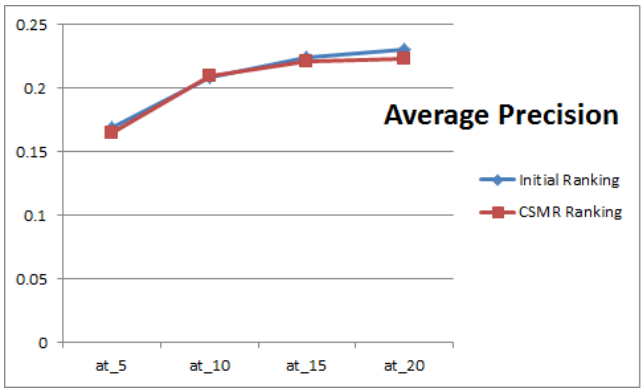

Figure 7: Average Precision scores for rankings at depth 5, 10, 15 and 20 for CSMR and our initial ranking algorithm (Indri).

evaluation metrics, even when applied on different ranking algorithms.

In the future, we intend to develop a novel click model which will incorporate the notion of document popularity. Moreover, we plan to implement and test our metric on other text similarity algorithms which use improved text similarity measures. Finally, performing experiments using massive amounts of click-through data obtained from commercial search engines, constitutes one of our major future plans, in order to obtain stronger confirmations about the well performance of our proposed evaluation metric.

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